



Trees 1 (Introduction)

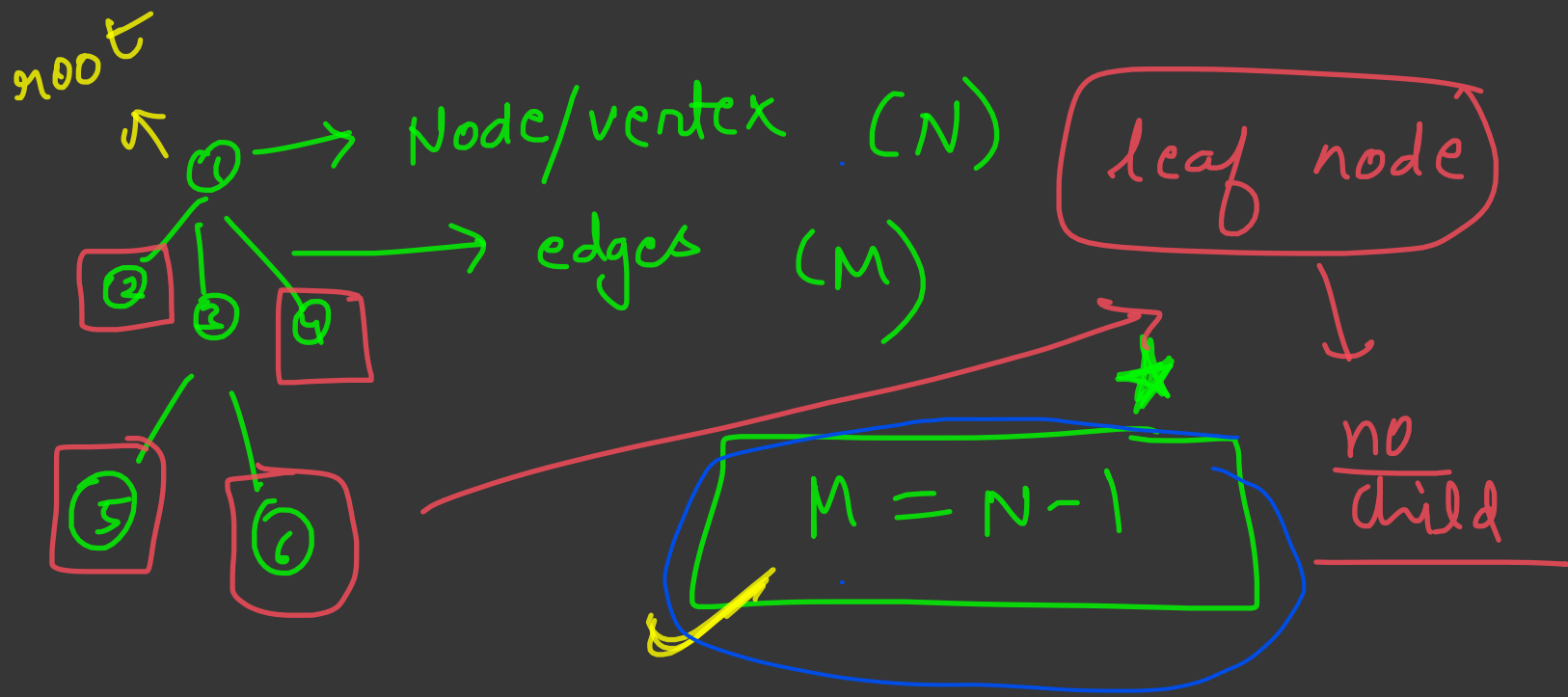
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5 star at codechef (2040)

Introduction to Trees

- Basic terminologies in trees
- Properties of trees
- How to store a tree? How will the input be given?
- Traversal techniques in Trees
 - Depth First Search (DFS)
 - Breadth First Search (BFS)
- Questions
 - Given a tree, give me the height of the tree
 - Subordinates
 - Given root node and another node x, print path root -> x



generic tree →

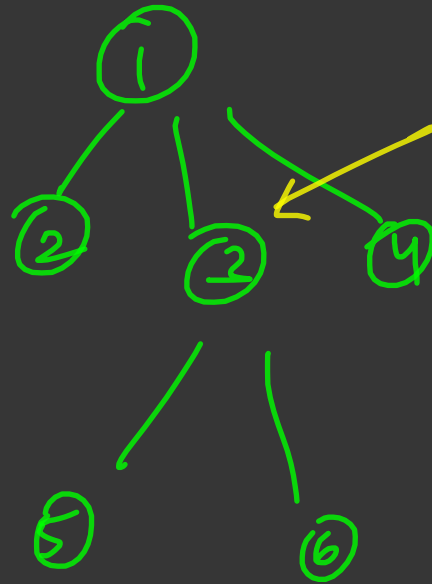


N -ary tree $\rightarrow$ each node can have n number of children

→ NO cycle in tree

Ancestor

node fall in path from root to the give node



children
→ 5, 6

parent
→ 1

Q

How to store a tree?

How will be the input given?

$N \rightarrow$ nodes

$u v$

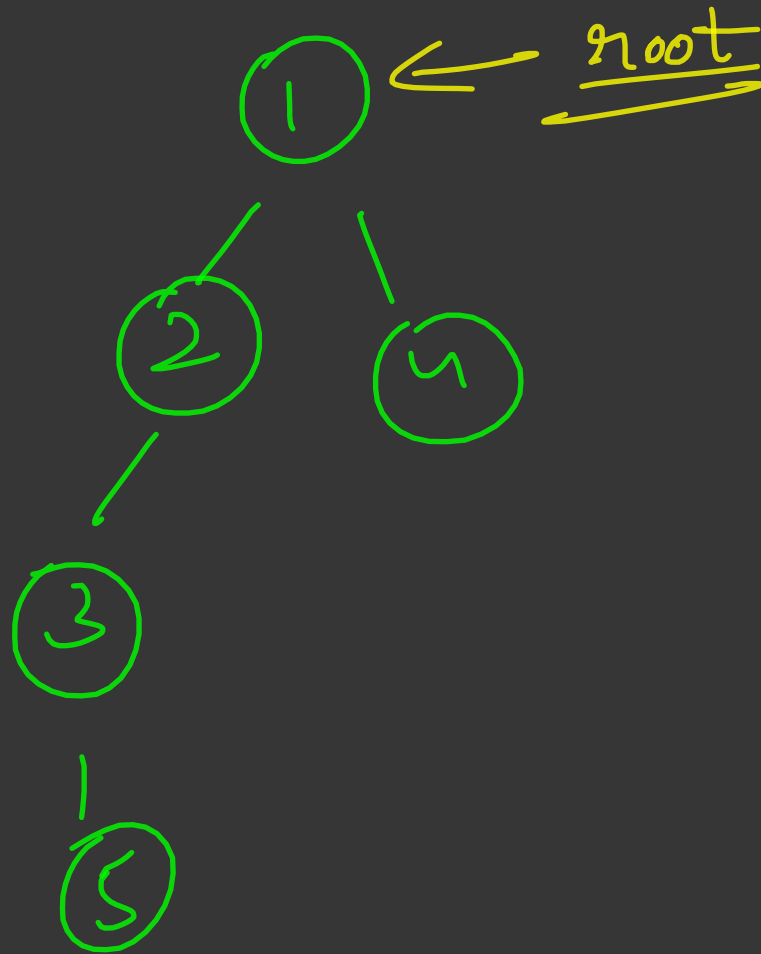
$N-1$ lines

tree given in this form
in questions

eg:

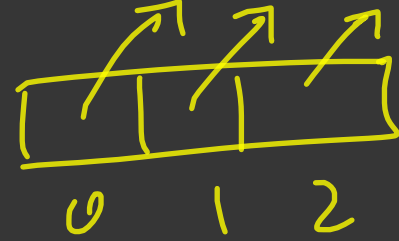
5
1 2
2 3
1 4
3 5

Tree can be
i) rooted=root given
ii) unrooted=root is not given
in that case we
assume 1 as root



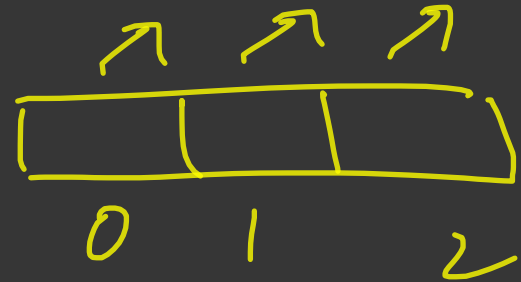
`int an[n+1],`

$n=2$



`vector<int> an[n+1];`

vector vector vector



how to store tree
in the form of adjacency list

square bracket hai
small bracket nahi isleye ye 2D hai 1D nahi

5

adj list

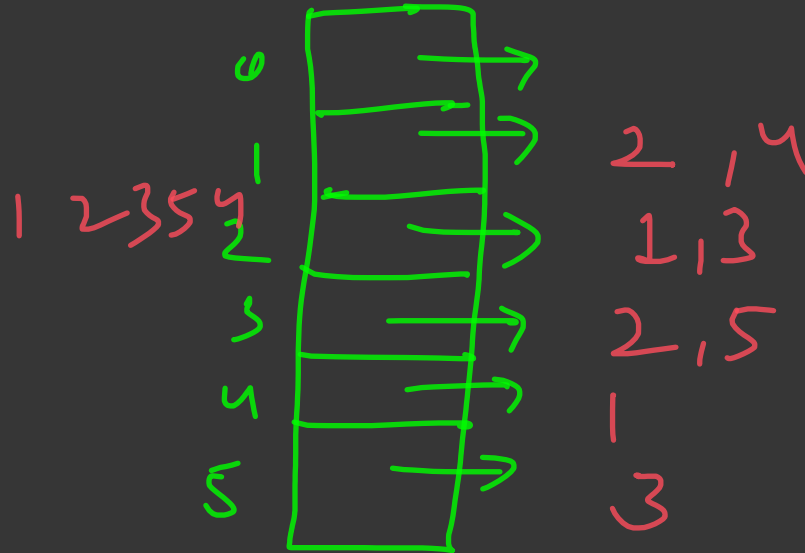
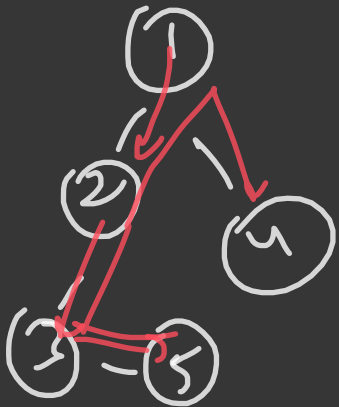
1 2

2 3

1 4

3 5

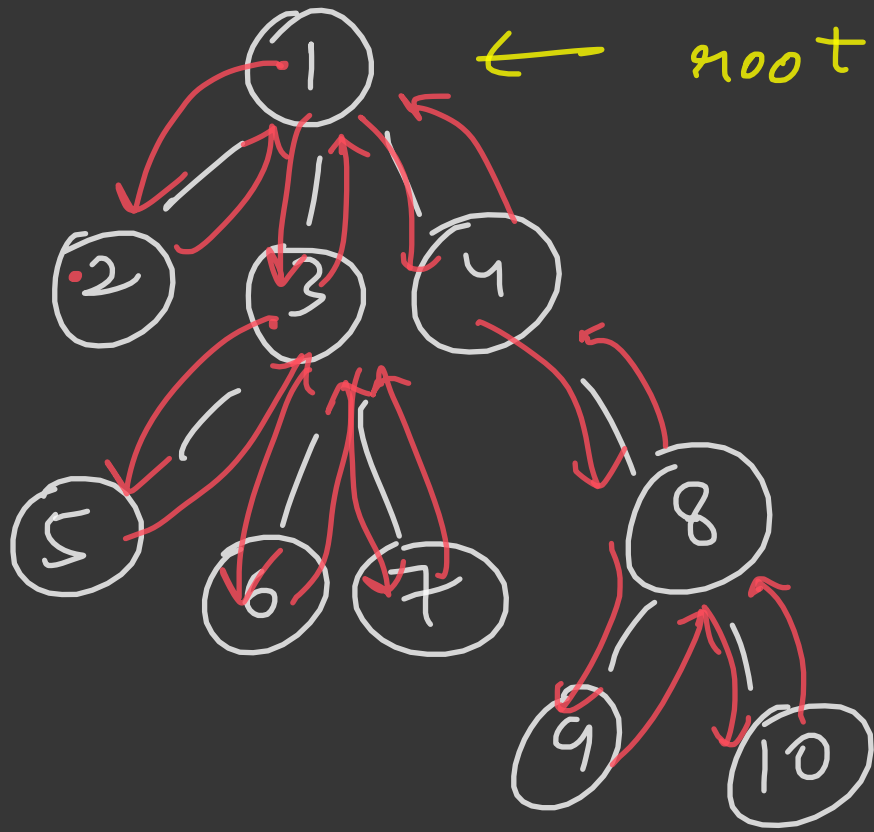
vector<int> adj[6]



Traversal → DFS → Depth first search
→ BFS → Breadth first search

① DFS → Depth first search

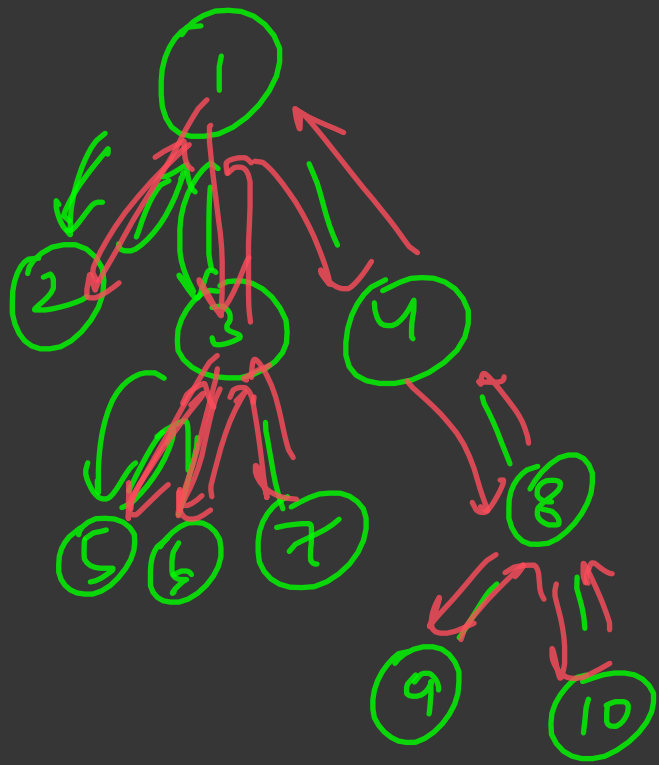
using parent while coding dfs so that child of a node don't move to it's parent back



→ code
can be
a recursive
one

~~node~~ $\rightarrow$ 2 adj

~~par~~ $\rightarrow$ 1

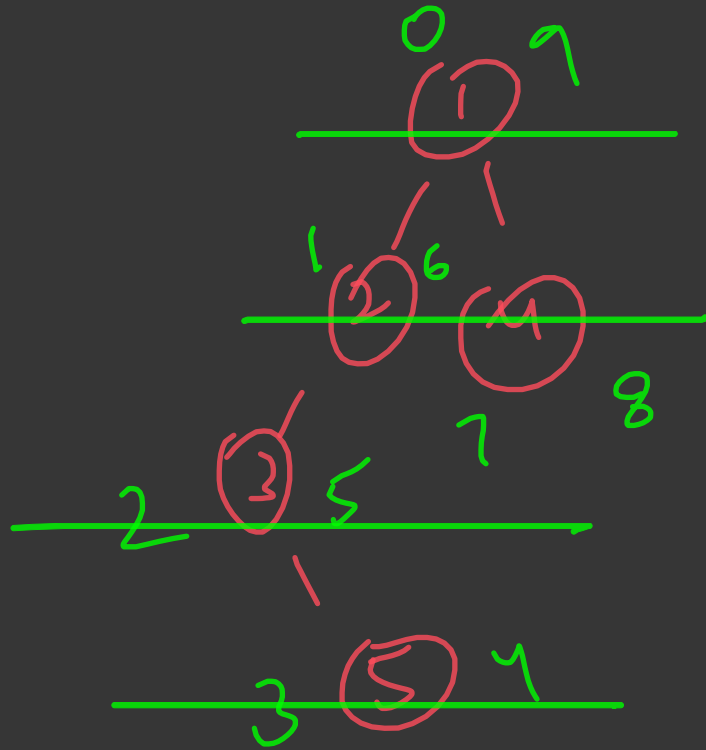


$\downarrow$
 $1 \rightarrow [2, 3, 4]$

$\downarrow$
 $3 \rightarrow [1, 5, 6, 7]$

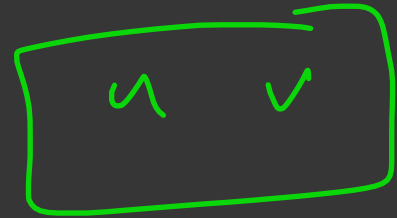
$5 \rightarrow [3]$

1 2 3 5 6 7 4 8 9 10



$n \rightarrow 10^5$
 $q \rightarrow 10^5$

9 queries



check if u is
anc. of v or
not

$$u \rightarrow 2$$

$$v \rightarrow 5$$



ancestor condition for (u,v)

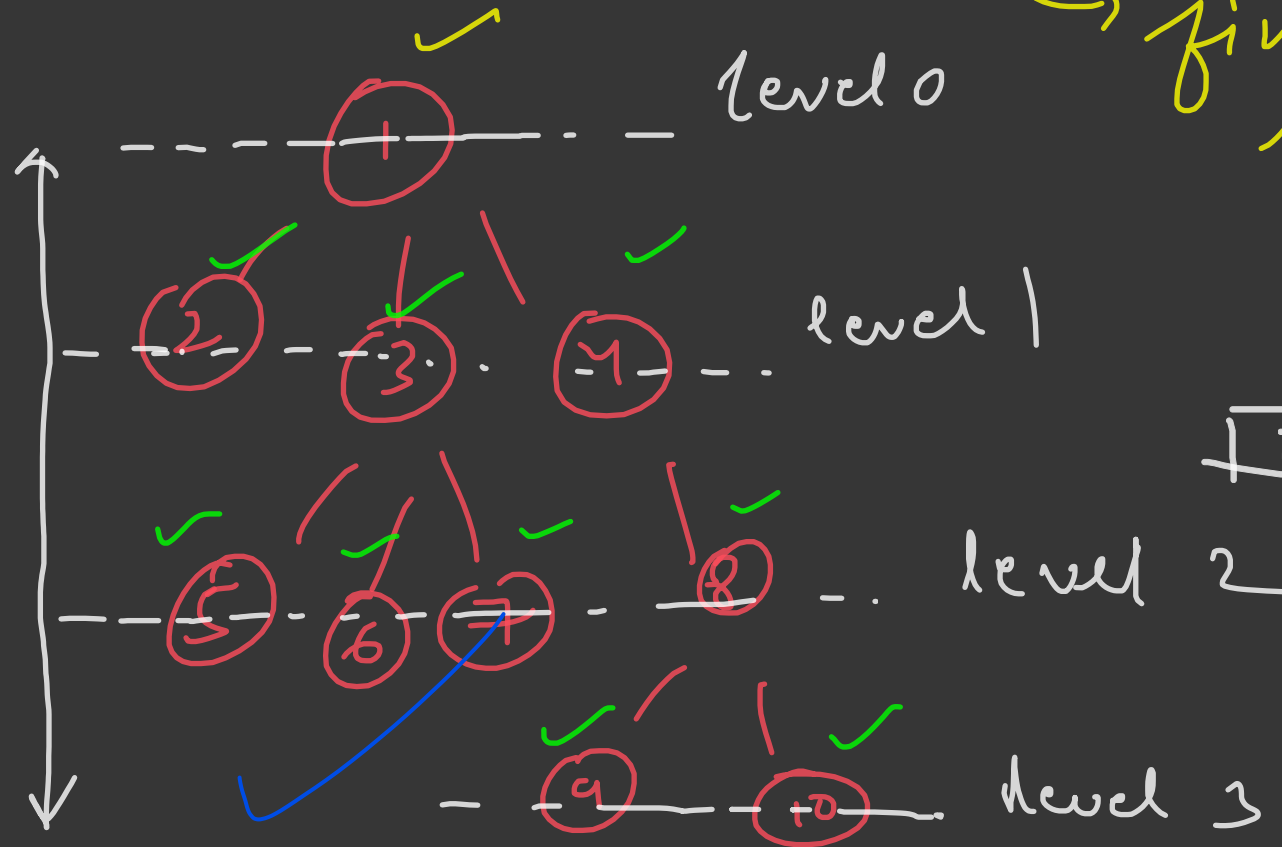


→

$$in_u < in_v < out_v < out_u$$

② BFS (Breadth first search)

finish one level and then move to next



1 2, 3, 4

①
② ③ ④
⑤ ⑥ ⑦ ⑧

9 10

Q given a tree, tell me the height of the tree.

cses subordinate problem

Q

$$\text{subtree}[4] = 1$$

$$\text{subtree}[2] = 2$$

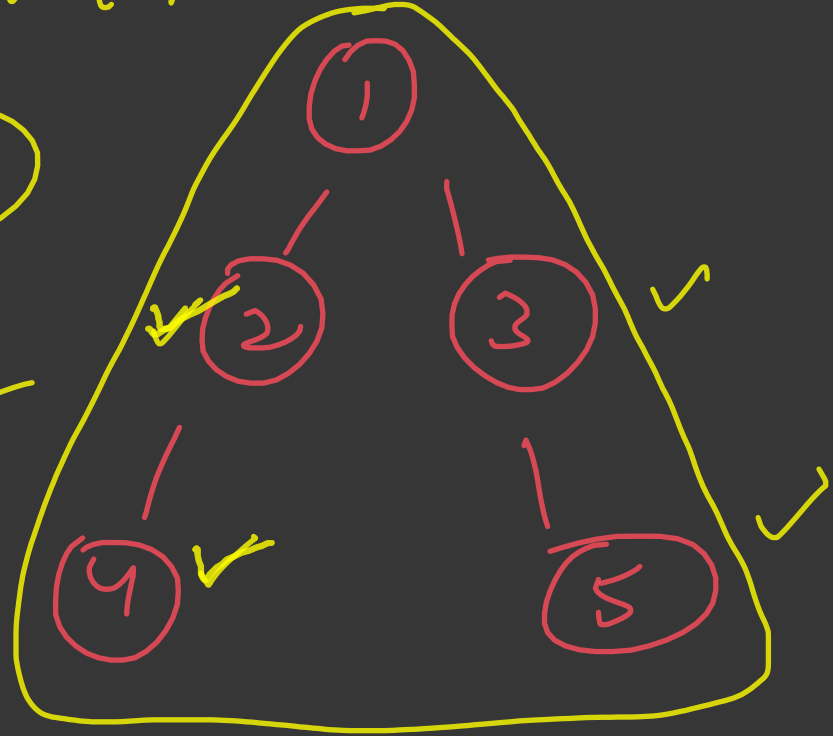
$$\text{subtree}[1] \rightarrow (5)$$

↓ 1 +

$$\text{subtree}[2] \rightarrow 2$$

+

$$\text{subtree}[3] \rightarrow 2$$



dp $\rightarrow$

	1	5	2	2	1	1
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0 1 2 3 4 5
4 1 1 0 0

